

Appendix: Comparative Tool Matrix I

Symbol key (applies to all cells): **Y** = full native support · ~ = partial or plugin-based · **N** = absent. See also Figure 4 for a colour-coded heatmap rendering of this table.

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Table 1: Comparative feature matrix (14 capabilities $\times$ 10 tools). Y = full native support, ~ = partial or plugin-based, N = absent.

Capability	template/ Snakemake	Nextflow	CWL	Quarto	Jupyter Book	R Mark- down	DVC	Overleaf	OpenAI Prism
Pipeline orchestration	Y	Y	Y	Y	~	N	N	Y	N
Manuscript rendering	Y	N	N	N	Y	Y	Y	N	Y
Testing enforcement	Y	N	N	N	N	N	N	N	N
Coverage thresholds	Y	N	N	N	N	N	N	N	N
Cryptographic provenance	Y	N	~ ¹	N	N	N	N	~ ²	N
Steganographic watermarking	Y	N	N	N	N	N	N	N	N
Multi-project management	Y	N	N	N	N	N	N	N	~
AI-agent documentation	Y	N	N	N	N	N	N	N	~
Agentic skill protocol (SKILL.md / MCP)	Y	N	N	N	N	N	N	N	N
Interactive TUI	Y	N	N	N	N	N	N	N	N
Zero-mock policy	Y	N	N	N	N	N	N	N	N
Container support	~	Y	Y	Y	N	N	N	N	N

Appendix: Comparative Tool Matrix II

Distributed execution	N	Y	Y	Y	N	N	N	~ ³	N	N
Multi-language (R/Julia)	N	Y	N	Y	Y	Y	Y	Y	N	N

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¹ Nextflow lineage records operate at workflow scope rather than as a rendered-document citation graph. ² DVC provides content-addressed versioning for data artifacts via its object store. ³ DVC integrates with remote storage (S3, GCS, Azure) but does not natively orchestrate distributed compute. This matrix is a versioned manuscript snapshot, not a continuously updated product survey. A future refresh should re-check every external capability against current primary documentation before changing a cell.